PKCERT Summer Internship 2026

Task 04 – NumPy & Pandas for Data Analysis

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Domain: AI & Software Development

Introduction

The objective of this task was to develop practical skills in NumPy and Pandas for numerical computing, data manipulation, and exploratory data analysis (EDA). NumPy was used to perform array operations and mathematical computations, while Pandas was used to organize, analyze, and visualize a public Loan Prediction dataset.

Part A – NumPy Fundamentals

Objective

To understand the fundamental concepts of NumPy and perform efficient numerical computations using arrays.

NumPy was used to create one-dimensional and multi-dimensional arrays using different methods. Array indexing, slicing, reshaping, and mathematical operations were performed to manipulate data efficiently. Broadcasting was demonstrated to perform operations on arrays of different shapes without manually resizing them, making the code faster and more efficient. Vectorized operations were compared with traditional loops, showing that vectorization improves performance and reduces code complexity. Linear algebra operations such as dot product, matrix multiplication, transpose, and inverse were also implemented.

Advantages of Broadcasting

- Reduces the amount of code.
- Improves execution speed.
- Uses memory efficiently.
- Simplifies mathematical operations on arrays.

Ouput:

```
PS C:\Users\Ayesha\Desktop\Task#04> cd "Part A"
PS C:\Users\Ayesha\Desktop\Task#04\Part A> py numpy_fundamentals.py
===== 1. Creating NumPy Arrays =====
1D Array:
[10 20 30 40 50]

2D Array:
[[1 2 3]
 [4 5 6]]

Array using arange():
[ 1  2  3  4  5  6  7  8  9 10]

Zeros Array:
[[0. 0. 0.]
 [0. 0. 0.]]

Ones Array:
[[1. 1.]
 [1. 1.]
 [1. 1.]]

Linspace Array:
[ 0.   2.5  5.   7.5 10. ]

===== 2. Indexing and Slicing =====
First Element: 10
Last Element: 50

First Three Elements:
[10 20 30]
```

```
===== 5. Broadcasting =====
Matrix:
[[1 2 3]
 [4 5 6]]

Vector:
[10 20 30]

Broadcasting Result:
[[11 22 33]
 [14 25 36]]

===== 6. Vectorized Operations =====
Loop Result:
[np.int64(2), np.int64(4), np.int64(6), np.int64(8), np.int64(10), np
np.int64(16), np.int64(18), np.int64(20)]

Vectorized Result:
[ 2  4  6  8 10 12 14 16 18 20]

===== 7. Linear Algebra =====
Matrix 1:
[[1 2]
 [3 4]]

Matrix 2:
[[5 6]
 [7 8]]

Dot Product:
[[19 22]
 [43 50]]
```

```
Second Row:
[4 5 6]

Element at Row 1 Column 2:
2

===== 3. Reshaping =====
Original Array:
[ 1  2  3  4  5  6  7  8  9 10 11 12]

Reshaped (3x4):
[[ 1  2  3  4]
 [ 5  6  7  8]
 [ 9 10 11 12]]

===== 4. Mathematical Operations =====
Addition:
[ 7 14 21]
Subtraction:
[3 6 9]
Multiplication:
[10 40 90]
Division:
[2.5 2.5 2.5]
Square:
[ 25 100 225]
Square Root:
[2.23606798 3.16227766 3.87298335]
```

```
Matrix Multiplication:
[[19 22]
 [43 50]]

Transpose:
[[1 3]
 [2 4]]

Inverse:
[[-2.   1. ]
 [ 1.5 -0.5]]
PS C:\Users\Ayesha\Desktop\Task#04\Part A>
```

Part B – Pandas Fundamentals

Objective

To learn how to create, manipulate, and analyze structured data using the Pandas library.

Pandas was used to create and manipulate Series and DataFrames. Indexing, filtering, sorting, and selection operations were performed to access and organize data. GroupBy operations summarized the dataset, while merge and join operations combined multiple DataFrames. These features simplify data preprocessing and analysis.

Output:

```
PS C:\Users\Ayesha\Desktop\Task#04\Part B> py pandas_fundamentals.py
===== 1. Creating Pandas Series =====
Ali       85
Ayesha    90
Ahmed     78
Sara      92
Fatima    88
dtype: int64

===== 2. Creating DataFrame =====
     Name  Age Department  Marks
0     Ali   20         AI     85
1  Ayesha   21         CS     90
2   Ahmed   22         AI     78
3    Sara   20         SE     92
4  Fatima   23         CS     88

===== 3. Indexing and Selection =====

Names Column:
0       Ali
1    Ayesha
2     Ahmed
3      Sara
4    Fatima
Name: Name, dtype: str

First Three Rows:
     Name  Age Department  Marks
0     Ali   20         AI     85
1  Ayesha   21         CS     90
2   Ahmed   22         AI     78

Row with Index 2:
Name          Ahmed
Age              22
Department       AI
Marks            78
Name: 2, dtype: object

Name and Marks Columns:
     Name  Marks
0     Ali     85
1  Ayesha     90
2   Ahmed     78
3    Sara     92
4  Fatima     88

===== 4. Filtering =====
Students with Marks greater than 85:
     Name  Age Department  Marks
1  Ayesha   21         CS     90
3    Sara   20         SE     92
4  Fatima   23         CS     88

===== 5. Sorting =====
Sorted by Marks (Highest to Lowest):
     Name  Age Department  Marks
3    Sara   20         SE     92
1  Ayesha   21         CS     90
4  Fatima   23         CS     88
0     Ali   20         AI     85
2   Ahmed   22         AI     78

===== 6. GroupBy =====
Average Marks by Department:
Department
AI    81.5
CS    89.0
SE    92.0
Name: Marks, dtype: float64

===== 7. Merging DataFrames =====
     Name  Age Department  Marks  Attendance
0     Ali   20         AI     85          90
1  Ayesha   21         CS     90          95
2   Ahmed   22         AI     78          85
3    Sara   20         SE     92          92
4  Fatima   23         CS     88          88

===== 8. Joining DataFrames =====
     Name  Age Department  Marks Grade
0     Ali   20         AI     85     B
1  Ayesha   21         CS     90     A
2   Ahmed   22         AI     78     C
3    Sara   20         SE     92     A
4  Fatima   23         CS     88     B
```

Part C – Data Analysis Mini Project

Objective

To perform exploratory data analysis on a public Loan Prediction dataset using Pandas.

A public Loan Prediction Dataset containing 45,000 records was analyzed using Pandas. The dataset was loaded from a CSV file, and missing values were checked using isnull().sum(). Since no missing values were found, no additional data cleaning was required. Summary statistics were generated using describe(), and exploratory data analysis was performed to examine applicant demographics, income, loan amount, credit score, and loan status. GroupBy, filtering, sorting, and Matplotlib visualizations were used to identify patterns and summarize the data.

Key Findings

- The dataset contains 45,000 loan application records.
- No missing values were found.
- GroupBy operations effectively summarized applicant information.
- Visualizations helped identify trends in income, education, loan amount, and loan status.

Output:

```
PS C:\Users\Ayesha\Desktop\Task#04\Part C> py data_analysis_project.py

First Five Rows
   person_age person_gender  ... previous_loan_defaults_on_file  loan_status
0          22        female  ...                             No            1
1          21        female  ...                            Yes            0
2          25        female  ...                             No            1
3          23        female  ...                             No            1
4          24          male  ...                             No            1

[5 rows x 14 columns]

Dataset Information
<class 'pandas.DataFrame'>
RangeIndex: 45000 entries, 0 to 44999
Data columns (total 14 columns):
 #   Column                          Non-Null Count  Dtype
---  ------                          --------------  -----
 0   person_age                      45000 non-null  int64
 1   person_gender                   45000 non-null  str
 2   person_education                45000 non-null  str
 3   person_income                   45000 non-null  int64
 4   person_emp_exp                  45000 non-null  int64
 5   person_home_ownership           45000 non-null  str
 6   loan_amnt                       45000 non-null  int64
 7   loan_intent                     45000 non-null  str
 8   loan_int_rate                   45000 non-null  float64
 9   loan_percent_income             45000 non-null  float64
 10  cb_person_cred_hist_length      45000 non-null  int64
 11  credit_score                    45000 non-null  int64
 12  previous_loan_defaults_on_file  45000 non-null  str
 13  loan_status                     45000 non-null  int64
dtypes: float64(2), int64(7), str(5)
```

```
memory usage: 4.8 MB

Dataset Shape
(45000, 14)

Column Names
Index(['person_age', 'person_gender', 'person_education', 'person_income',
       'person_emp_exp', 'person_home_ownership', 'loan_amnt', 'loan_intent',
       'loan_int_rate', 'loan_percent_income', 'cb_person_cred_hist_length',
       'credit_score', 'previous_loan_defaults_on_file', 'loan_status'],
      dtype='str')

Missing Values
person_age                        0
person_gender                     0
person_education                  0
person_income                     0
person_emp_exp                    0
person_home_ownership             0
loan_amnt                         0
loan_intent                       0
loan_int_rate                     0
loan_percent_income               0
cb_person_cred_hist_length        0
credit_score                      0
previous_loan_defaults_on_file    0
loan_status                       0
dtype: int64

No missing values found in the dataset.
```

```
Summary Statistics
         person_age  person_income  ...  credit_score   loan_status
count  45000.000000   4.500000e+04  ...  45000.000000  45000.000000
mean      27.764178   8.031905e+04  ...    632.608756      0.222222
std        6.045108   8.042250e+04  ...     50.435865      0.415744
min       20.000000   8.000000e+03  ...    390.000000      0.000000
25%       24.000000   4.720400e+04  ...    601.000000      0.000000
50%       26.000000   6.704800e+04  ...    640.000000      0.000000
75%       30.000000   9.578925e+04  ...    670.000000      0.000000
max      144.000000   7.200766e+06  ...    850.000000      1.000000

[8 rows x 9 columns]

Gender Distribution
person_gender
male      24841
female    20159
Name: count, dtype: int64

Education Distribution
person_education
Bachelor       13399
Associate      12028
High School    11972
Master          6980
Doctorate        621
Name: count, dtype: int64
```

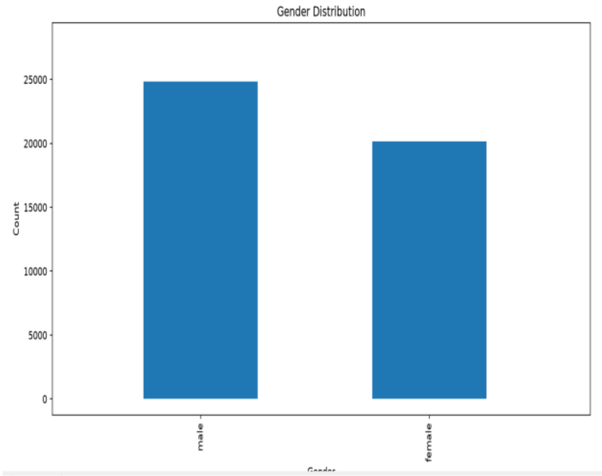
```
Home Ownership Distribution
person_home_ownership
RENT        23443
MORTGAGE    18489
OWN          2951
OTHER         117
Name: count, dtype: int64

Loan Intent Distribution
loan_intent
EDUCATION            9153
MEDICAL              8548
VENTURE              7819
PERSONAL             7552
DEBTCONSOLIDATION    7145
HOMEIMPROVEMENT      4783
Name: count, dtype: int64

Loan Status
loan_status
0    35000
1    10000
Name: count, dtype: int64

Average Person Age
27.764177777777778

Average Person Income
80319.05322222222

Average Loan Amount
9583.157555555556
```

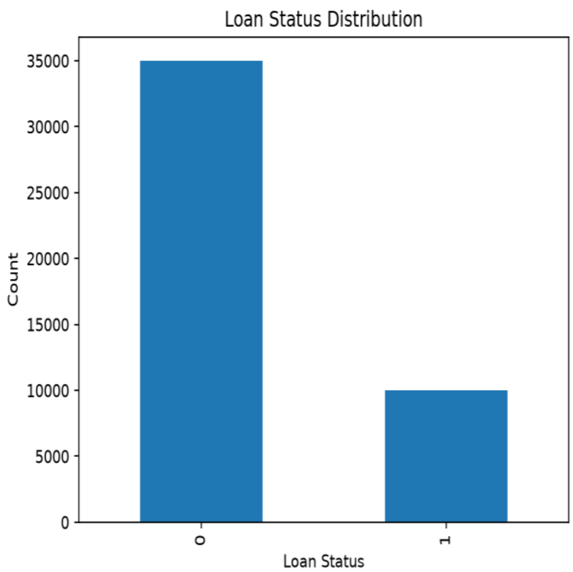
```
Average Credit Score
632.6087555555556

Average Income by Gender
person_gender
female    79410.904856
male      81056.034942
Name: person_income, dtype: float64

Average Loan Amount by Education
person_education
Associate      9627.988942
Bachelor       9556.113068
Doctorate      9930.932367
High School    9543.422987
Master         9595.030229
Name: loan_amnt, dtype: float64

Average Credit Score by Home Ownership
person_home_ownership
MORTGAGE    633.036779
OTHER       627.803419
OWN         632.058285
RENT        632.364458
Name: credit_score, dtype: float64
```

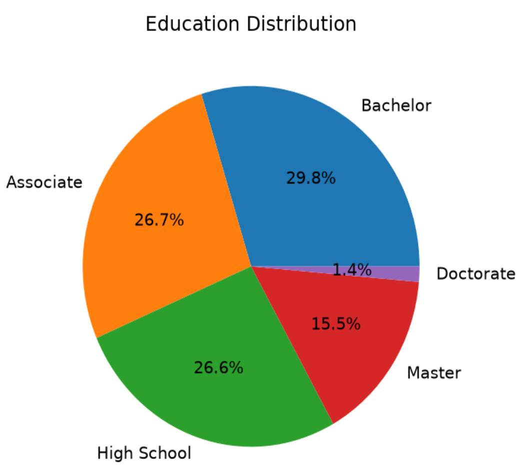
```
Loan Status by Loan Intent
loan_intent        loan_status
DEBTCONSOLIDATION  0              4982
                   1              2163
EDUCATION          0              7601
                   1              1552
HOMEIMPROVEMENT    0              3525
                   1              1258
MEDICAL            0              6170
                   1              2378
PERSONAL           0              6031
                   1              1521
VENTURE            0              6691
                   1              1128
Name: count, dtype: int64

Applicants with Income Greater Than 100000
       person_age  person_income  loan_amnt
8              24         100684      35000
10             22         102985      35000
12             23         114860      35000
13             26         130713      35000
14             23         138998      35000
...           ...            ...        ...
44947          42         705960      20686
44965          31         100407      17783
44972          43         184295      25000
44982          26         130495       7622
44990          31         136832      12319

[10024 rows x 3 columns]
```

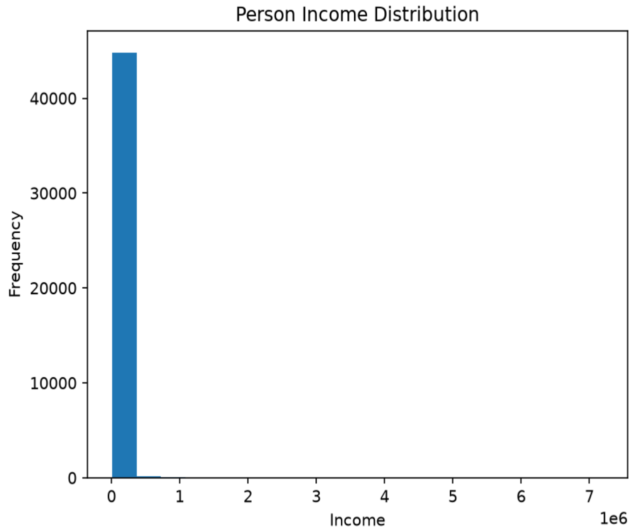
```
Applicants with Loan Amount Greater Than 15000
       person_age  loan_amnt        loan_intent
0              22      35000           PERSONAL
3              23      35000            MEDICAL
4              24      35000            MEDICAL
6              26      35000          EDUCATION
7              24      35000            MEDICAL
...           ...        ...                ...
44975          23      16474            MEDICAL
44983          30      16000  DEBTCONSOLIDATION
44984          24      20000            MEDICAL
44985          36      18000           PERSONAL
44986          34      19594  DEBTCONSOLIDATION

[6855 rows x 3 columns]

Top 10 Highest Income Applicants
       person_age person_gender  ... previous_loan_defaults_on_file  loan_status
32297         144        female  ...                             No            0
38113         109          male  ...                             No            0
37930         116          male  ...                             No            0
30049          42          male  ...                            Yes            0
32546          60        female  ...                            Yes            0
32497          63          male  ...                             No            0
37175          42          male  ...                             No            0
41288          46          male  ...                            Yes            0
31924          44          male  ...                            Yes            0
35850          47          male  ...                            Yes            0

[10 rows x 14 columns]

Data Analysis Completed Successfully!
```

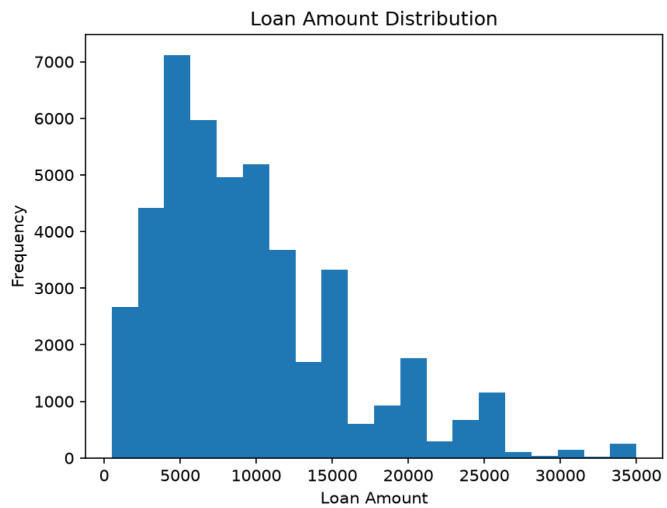
Graphs:


Gender Distribution
25000
20000
15000
10000
5000
0
Count
male
female


Loan Status Distribution
35000
30000
25000
20000
15000
10000
5000
0
Count
0
1
Loan Status


Education Distribution
Bachelor
29.8%
Associate
26.7%
1.4%
Doctorate
15.5%
Master
26.6%
High School





Conclusion

This task provided practical experience with NumPy and Pandas for numerical computing and data analysis. It improved understanding of array operations, data manipulation, exploratory data analysis, and visualization techniques, which are essential skills for artificial intelligence and data science.

GitHub Repository Link:

https://github.com/AyeshaBhattiML/PKCERT-AI-Task-04.git